

Multiple Choice

1. **Answer:** B

Options: A) $P(E, F), P(H), P(E|H), P(F|H)$; B) $P(E, F), P(H), P(E, F|H)$; C) $P(H), P(E|H), P(F|H)$; D) $P(E, F), P(E|H), P(F|H)$

Note: Correct option: B - $P(E, F), P(H), P(E, F|H)$

2. **Answer:** D

Options: A) 3; B) 4; C) 7; D) 15

Note: Correct option: D - 15

3. **Answer:** B

Options: A) Larger if the error rate is larger.; B) Larger if the error rate is smaller.; C) Smaller if the error rate is smaller.; D) It does not matter.

Note: Correct option: B - Larger if the error rate is smaller.

4. **Answer:** A

Options: A) $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$; B) $P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)$; C) $P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)$; D) $P(X, Y, Z) = P(X) * P(Y) * P(Z)$

Note: Correct option: A - $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$

5. **Answer:** A

Options: A) Decreases model bias; B) Decreases estimation bias; C) Decreases variance; D) Doesn't affect bias and variance

Note: Correct option: A - Decreases model bias

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'lower'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Can use a mix of different activation functions'.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $P(A, B | C) * P(C)$ '.

Long Form Response

11. **Answer:** convolutional networks. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Unsupervised learning. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key